REItrades V1.2 — Work Breakdown & Estimate

Prepared for: Artur Muller, Founder & CEO, REItrades **Engineer:** Brian Takita — senior full-stack (Rust / Vue / Postgres) **Rate:** $100/hr **Basis:** Itemized from your V1.2 feature brief and the Listing Creation Steps PDF, priced as **agentic-implementation hours** — one senior AI-native engineer using coding agents for search, scaffolding, migration/test drafts, and parallel review.

1. How to read this estimate

- **These are agentic hours — compressed by an AI-native workflow.** They already assume heavy use of coding agents, so you get senior-quality delivery faster and at a lower total than a conventional solo build. They still include human review, integration, targeted tests, and production QA on a **live** marketplace — the parts agents do not compress.
- **Ranges are real.** The low end applies when your existing patterns are clean and reusable; the high end when data models are inconsistent or production-migration risk is high. My AI-native workflow biases delivery toward the **low end of each range**, so you're most likely to land there — from efficiency, not from cutting scope.
- **Rows are incremental within a shared implementation.** Admin grid columns/filters/inline-edit, bulk-action plumbing, audit logging, import/export, role gates, and listing-data mapping are built **once** and reused. Per-row numbers assume those primitives already exist, so they do not sum naively — see the de-duped roll-up.
- **What agents compress vs. what they don't:**
  - Admin table/UI work: ~25-40% faster.
  - Rust/Postgres permission & data-model work: ~15-30% faster.
  - External integrations, production migrations, PDF/OM output, AI extraction accuracy: **least** — validation dominates.
- **"100% accurate" is scoped as a deterministic core with an AI assist** (see §3b), **not** a pure-AI promise. Deterministic code owns every number, validation rule, structured-data lookup, and document assembly — those are *provably* correct. The model only *proposes* values for the genuinely fuzzy work (extraction from messy source docs, generated narrative), and every proposal passes deterministic validators + confidence scoring + the A–D provenance rubric (§3a) + admin review before it ships. I will not present AI extraction as inherently 100% accurate without that verification path, especially for buyer-facing materials.

2. Scope at a glance

Three workstreams against a launched (post-beta, 10/14/2025) production platform:

Workstream	What it covers
1. Admin DBMS	Listings/user/company management: columns, inline edit, filters, bulk actions, MailerLite status, import/export, audit trails, deletes, NDA/LOI visibility, activity analytics.
2. Broker segment	New account type, application + approval workflow, separate broker admin views, role-gated pages/permissions, per-listing access codes, Broker/Principal tags.
3. Listing creation + OM rebuild	Rust/Tera rebuild, step gating, asset-class schemas, teaser/Offering Memorandum (OM) data derivation + production migration, AI/BatchData marketing materials, OM polish, broker OM upload, old-OM extraction, Stripe listing tiers.

3. Itemized task breakdown

0. Setup / technical audit

Task	Hours	Notes
Repo/data-model audit, route/module map, milestone plan	16-28	Agents parallelize codebase mapping; final architecture & sequencing need senior review.
Production-safety plan for live beta data	8-16	Migrations, rollback, delete policy, regression-risk mapping.
Subtotal	**24-44**	

*This Phase-0 discovery work is **credited toward Phase 1** if we proceed — it is not an extra charge on top of the build, and it firms up every downstream estimate before any production code is written.*

1.1 Listings Management

Task	Hours	Notes
Delete listing from action dropdown; cascade related data	16-28	A bare soft-delete flag is ~4-8h; the 16-28h is the **cascade** — safely handling related NDAs/LOIs/files/analytics + cleanup jobs, audit trail, and tests against live production data. Drops to the low end if scope is a simple archive without cascade cleanup.
View NDAs and LOIs signed on each listing	8-16	Mostly query/UI if documents are stored consistently.

1.2 User Management

Task	Hours	Notes
Update user summary-table columns exactly as listed	10-18	Add/remove columns, enum/multi-select display, sort.

Task	Hours	Notes
View/edit complete user profile pages (incl. analytics)	14-26	More if user/contact/company data is split across inconsistent models.
Email status Active / Not Active / bounced from MailerLite	10-20	Depends on MailerLite webhook/API access.
Search/filter users by all data points	12-24	Needs query-builder/index review so production filters stay fast.
Inline edit on summary table (no dropdown/popup)	16-30	Save/validation/error states need careful review.
50 contacts per page	2-4	Small if pagination exists.
Bulk platform invitations to selected users	10-18	Selection state, confirmation, background job/throttle, audit.
Send teasers/flyers to selected users from user mgmt	14-28	Reuses teaser path with new recipient semantics; must prevent accidental broad sends.
Bulk import/export users with audit trail	18-34	CSV/schema validation, dry-run/errors, audit records.
Remove "Invite network" button	1-2	UI removal + route/action check.
New bulk-actions button for selected users	6-10	Menu/actions plumbing reusing bulk framework.
Delete users individually and in bulk	10-20	Soft-delete policy, account-access effects, company/contact relations.

1.2 User Analytics / Activity

Task	Hours	Notes
Last signed in	3-8	Small if auth/event table records it.
NDAs signed through platform	4-8	Relationship display if NDA records normalized.
LOIs submitted through platform	4-8	Same pattern as NDA.
Property-teaser engagement from MailerLite	14-28	Harder — depends on campaign/message-ID mapping to MailerLite engagement API.
History of messages sent through platform	8-18	Depends on whether outbound sends are centrally logged.

1.3 Company Management

Task	Hours	Notes
Update company summary-table columns exactly as listed	8-16	Simpler than user table.

Task	Hours	Notes
View/edit full company profiles	12-22	Edit form, validation, related records.
View users/contacts per company	6-12	Straightforward if every user has a company relation.
Search/filter companies	10-20	Reuse user-management filter/query primitives.
50 companies per page	2-4	Small if pagination exists.
Bulk import/export companies with audit trail	16-30	Slightly simpler than user import/export.
Delete company individually and in bulk	10-20	Define behavior for attached contacts/users first.
New bulk-actions button for companies	4-8	Reuse bulk-action menu pattern.

2.1 Broker Database

Task	Hours	Notes
Separate broker DB management under admin	22-40	Assumes logical separation in same Postgres app. **Physical separate DB/schema: +16-40.**
Broker user table (Name/Title/Email/Company/ License#/State/LinkedIn/Activated/Created/Email Status)	6-12	Incremental after admin-table primitive.
Broker company table (Name/License#/State/Website/ LinkedIn/Contact Count)	6-12	Same pattern as company admin table.

2.2 Approval Workflow

Task	Hours	Notes
Broker applies free, shows CRE experience	8-16	Application fields + validation.
Broker application flow	16-30	Public/app flow, persistence, admin visibility, notifications.
Approval gating before platform entry	14-28	Role/status gating, route guards, denied-access tests.

2.3 Broker Pages / Permissions

Task	Hours	Notes
Broker pages/capabilities differ from principals	10-18	Shared role-gate framework; removals below are incremental.
Remove User Acquisition Criteria box (brokers)	2-6	Keep code-added listings.
Remove Browse page for brokers	6-12	Nav guard + direct-URL protection.

Task	Hours	Notes
Remove Browse/Acquisition/Request-BOV buttons from broker dashboard	2-6	UI + server-side permission check.
Engaged Listings = code-accessed + NDA-signed	10-20	Event/source-of-truth query.
Saved Listings includes code-accessed listings	8-16	Depends on saved/accessed data model.
Remove Company Peers (broker profile)	2-5	Small UI/route change.
Remove Company Activities box (broker profile)	2-5	Small UI/route change.
Per-listing access code, any logged-in user incl. brokers	18-34	Code generation, uniqueness, access rules, audit, anti-guessing protections.
Broker tag / Principal tag on listings	6-12	Data field, UI badges, optional filters.

3. Listing Creation & OM Rebuild (first asset class)

Confirmed wizard scope (from the V1.2 PDF "Listing Creation Steps" screenshots). The current flow is a **5-step wizard** with a **separate 5-state listing lifecycle** running alongside it:

- **Wizard steps:** (1) **Location** — asset class, address, city, state/province, postal code, entrance-location details; (2) **Property Overview** — building name, asset-class sub-category, main use, submarket, brand/flag, year built/renovated, total square footage, and asset-class-specific fields (the hotel example shows Number of Keys, Room size, Parking, Amenities); (3) **Financials** — an "Investment Snapshot" with ~16 fields (Asking Price, Cap Rate, Historical Occupancy, ADR, RevPAR, Gross Income, Vacancy, Effective Gross Income, Opex, NOI, LTV, Interest Rate, Amortization Period, Annual Debt Service); (4) **Attachments** — categorized document upload (Expenses Breakdown, CapEx, Rent Roll, Property Video) plus additional documents and a property-photo gallery, all drag-and-drop; (5) **Marketing Materials** — generated teaser/OM output.
- **Lifecycle status machine:** Draft → Under Review → Active → Pending → Closed, with the seller only able to submit to **Under Review** until REItrades approves (matches the brief's gating rule).
- **Per-step custom attributes:** every step exposes an "+ Add Attribute" control.

What this confirms for the estimate: the field density is real and asset-class-driven (the hotel example alone carries ~12 property-overview + ~16 financial fields, several hotel-specific), so the per-asset-class schema work sits at the **mid-to-high end** of the ranges below rather than the low end. It also surfaces one item the markdown-only pass under-counted — the categorized **Attachments/media step** — now broken out explicitly. The screenshots **validate** the prior ranges; they do not blow up the top line.

Task	Hours	Notes
Reimplement 5-step listing create/edit wizard in Rust/Tera, match current UI, consolidate modules	40-72	Core rebuild before feature bullets; 5 steps (Location, Property Overview, Financials, Attachments, Marketing Materials), visual parity + backend structure need review.

Task	Hours	Notes
Step locking + 5-state lifecycle (Draft→Under Review→Active→Pending→Closed) + Under-Review submission gating	22-40	Two coupled state machines: per-step unlock gating and permission-gated listing status transitions.
Asset-class required/optional + custom attributes (first class)	30-58	Field-dense per screenshots (~28 fields for the hotel example). **+12-24 per additional asset class** by field complexity.
Attachments step: categorized doc upload (Expenses/ CapEx/Rent Roll/Video) + additional docs + photo gallery, drag-and-drop	14-24	Newly broken out from the PDF; upload categories, validation, gallery ordering, storage wiring.
Teaser/OM data derivation + **production migration**	30-65	Major production-data risk; migration script + regression checks vs. existing listings.
Marketing Materials from BatchData + AI fallback; generate-once / regenerate	26-55	Idempotency, source tracking, failure states, manual regeneration.
Modular one-asset-class-at-a-time PR structure	8-14	Process/architecture overhead that lowers later review risk.

3. Additional Comments / Requests

Task	Hours	Notes
Improve AI-generated content (demographics, data accuracy)	32-75	Source-backed + validation/confidence/admin review, **graded against the accuracy rubric below** (§3a) — replaces the unmeasurable "100% accurate" ask with a defined acceptance bar.
Polish OM generator output	20-50	Depends on design expectations and PDF/rendering stack.
Ensure additional attributes flow into generated OM	6-14	Small if listing-data→OM mapping is clean.
Broker-only OM upload, NDA/ email workflow compatible; principals use generator	14-30	Permission, file handling, NDA-triggered delivery, attachment/link rules.
Old OM/flyer upload → extract → autofill listing + OM	45-100	**Largest AI item:** document ingestion, OCR/PDF parsing, schema extraction, confidence flags, review UI, sample-doc tests scored on the **field-accuracy rubric** (§3a).
AI accuracy rubric + evaluation harness	14-28	New line item (per "include rubrics in AI accuracy work"): per-field grading rubric, a labeled ground-truth sample set, and an automated scoring harness that gates release of extraction/generation changes. This is what makes "accurate" measurable and defensible to buyers.
3 listing tiers (Free / Premium $199 / Elite $499) as **new first step** before Location; Stripe links	18-38	The tier step is inserted ahead of the confirmed 5-step wizard, so it also shifts step-index/gating logic. Tier selection, state persistence, Stripe link flow, entitlement behavior, admin visibility.

3a. AI accuracy rubrics (acceptance criteria, not a "100% accurate" promise)

Every AI-touched field — extracted from an uploaded OM/flyer, or generated (demographics, marketing copy) — is graded on a **provenance rubric** that decides whether it can auto-publish or must be reviewed:

Grade	Meaning	Handling
A — Source-exact	Value copied verbatim from an authoritative source (BatchData, uploaded rent roll/financials, structured listing field)	Auto-fill, no review
B — Derived-verified	Computed from source fields by a deterministic rule (e.g. NOI = EGI – Opex) and reconciles within tolerance	Auto-fill, flagged for spot-check
C — AI-inferred	Produced by the model without a directly citable source	**Blocked from external send** until an admin approves; shown with its confidence and rationale
D — Missing/ low-confidence	No source and confidence below threshold	Requires manual entry; never auto-published

Acceptance thresholds (the "done" bar for an AI change, measured on the labeled sample set):

Field class	Target precision	Notes
Financial (Asking Price, Cap Rate, NOI, Opex, LTV…)	**≥ 99%** on Grade A/B; zero silent Grade-C financials sent externally	Money fields are buyer-facing; wrong numbers are the worst failure mode
Structured facts (address, sq ft, year built, keys/units)	**≥ 97%**	Mostly source-exact
Demographics / market data	**≥ 95%**, every value carrying a cited source	"much better demographics" from the brief, made testable
Narrative/marketing copy	Human-approved before external send	Graded for factual claims only, not style

The harness runs on a labeled ground-truth set (a handful of representative OMs per asset class), scores each release of the extraction/generation code, and **gates merge** on the thresholds above. That turns Artur's "100% accurate" request into a concrete, auditable acceptance test instead of an open-ended promise — and gives admin review a queue driven by grade, not guesswork.

Manual review workflow (the human-in-the-loop step). This is a first-class feature, not an implied side effect of the rubric: every AI-touched field carries a **confidence score**, and low-confidence readings (Grade C/D, or any value below its calibrated threshold) are **flagged and routed to a review queue instead of auto-publishing**. There are two review points, and they layer:

Review step	Who does it	What it catches
Listing-owner review	The seller/broker who uploaded the OM, at data-entry time, before submitting the listing to *Under Review*	Confirms/corrects autofilled fields — flagged low-confidence ones surfaced first — by the party who actually knows the property. Scales, because the effort sits with the owner.
Admin review gate	REItrades staff, before anything goes external	Nothing money-facing or buyer-facing (asking price, cap rate, NOI, teaser/OM content) ships until a flagged field is cleared. The trust backstop.

The flag is only trustworthy if the confidence score is **calibrated** — a value reported at 0.9 confidence should be right ~90% of the time — which is exactly what the eval harness above measures and tunes. Calibrated confidence + a graded review queue is what lets the platform show buyers "verified" data while keeping a human in the loop precisely where the model is unsure, and no wider.

3b. Deterministic core vs. AI assist (where "100%" is honest)

The strongest, most defensible version of the accuracy story is to split the pipeline into a **deterministic layer** and a **generative layer**, and let deterministic code own everything it possibly can. Then "100% accurate" attaches to the parts that can actually carry it, and the fuzzy parts are described as *validated + reviewed* — a claim that holds up on a call and in a contract.

Deterministic software is provably correct — this is where "100%" is literal:

Concern	Why it's deterministic (100%)
Calculations — cap rate, NOI, price/unit, price/SF, GRM, loan sizing, LTV	Formulas, not predictions. Computed in code from validated inputs; the model never does arithmetic.
Schema / field validation — types, ranges, required fields, unit normalization, enum membership (asset class, state, zoning)	A validator passes or fails; correct by construction.
Structured data retrieval — reading a known field (year built, SF, ownership) from BatchData / rent rolls / structured listing fields	Deterministic lookup. Only *locating* a value inside a messy source doc is fuzzy.
Template / OM rendering — Tera templates	Identical inputs produce identical output. Document assembly is deterministic; only the prose inside is generated.

The AI layer only proposes — it never publishes unchecked. For the genuinely probabilistic work — extraction from unstructured source (old OM PDFs, flyers, scanned rent rolls) and generated narrative (listing copy, market/demographic write-ups) — the model *proposes* values, and every proposal runs the gauntlet already defined above:

1. Deterministic code owns every number, validation rule, and document assembly — provably correct and auditable.
2. The model only proposes values for fuzzy extraction/generation.
3. Every model output passes deterministic validators + confidence scoring + the A–D provenance rubric (§3a) before it is shown.
4. Anything money-facing or buyer-facing gets admin review before it goes out.

Why lead with this: it is the strongest trust argument, it is cheaper and more testable than trying to make the AI itself infallible, and it fits a bootstrapped budget — deterministic checks are hours of code, not an open-ended model-tuning bill. If a *learned prediction* on tabular data is wanted later (valuation/cap-rate estimate, lead scoring, asset-class classification), that is a phase-2, data-dependent option requiring a labeled outcome set — and the right tool there is gradient-boosted trees (XGBoost/LightGBM), which beat plain neural nets on tabular data. It is out of scope for V1.2.

3c. Build vs. buy — where vendors lower cost

No document-extraction vendor is "100% accurate" on messy CRE source docs, so a vendor never replaces the deterministic core and the §3a/§3b verification path — it slots in as the *propose* layer that the validators and rubric then check. Within that frame, buying the right pieces meaningfully lowers build hours on the two most expensive, least agent-compressible items (old-OM extraction 45-100h; demographics/data accuracy 32-75h). The trade to weigh openly: every "buy" converts build hours into a recurring per-use fee — good for a bootstrapped budget upfront, but a standing cost to name.

Document extraction — three approaches (pick on a bake-off against your real OMs, not on vendor accuracy claims):

Approach	What you buy	Best for	Trade-off
Parser API → our schema (e.g. Reducto, Azure Document Intelligence, Google Document AI, AWS Textract, Landing AI)	Best raw table/layout accuracy on financials + bounding-box **citations** that map directly to Grade-A "source-exact" provenance	Money-facing fields that need a cited source	You still own schema mapping + validators
LLM-native structured extraction (Claude / Gemini with structured outputs)	One vendor for extraction *and* generation; adapts to new asset classes with no retraining	The long tail of odd layouts + fuzzy fields	Needs our validator/confidence layer; the model never does arithmetic (§3b)
Managed IDP with review queue (e.g. Rossum, Docsumo, Nanonets)	Extraction **plus** a built-in human-review UI + confidence + correction loop	Cutting the most upfront build hours	Highest recurring cost, least control over schema/grading

Recommended default: chain **parser → LLM → deterministic validators** — accurate tables with citations for provenance, an LLM for the fuzzy fields, our validators and the A–D rubric gating what ships. Keep the managed-IDP option as the lever if lowering upfront hours matters more than control.

Where vendors also reduce cost elsewhere:

- **Demographics / market data (biggest win).** Buy source-backed data — US Census/ACS (free), Esri, Attom, Precisely, Regrid — instead of AI-*generating* it. That turns demographics from Grade-C "AI-inferred" into Grade-A "source-exact": cheaper *and* more accurate at once, directly shrinking the demographics-accuracy item.
- **Address / geocode validation** (Smarty, Google, Mapbox) — deterministic normalization feeding the schema-validation layer as Grade-A.
- **Payments/tiers** — Stripe Payment Links / hosted pricing table (already Stripe in the brief): mostly configuration, not a custom checkout build.
- **Email status + engagement** — MailerLite's own API/webhooks already expose both; buy the tracking rather than build it.
- **OM/PDF rendering** — template-PDF vendors exist, but Tera already renders deterministically for free; only worth it if the cost there is design polish, not logic. Recommend keeping Tera.

How to decide without guessing: a short **extraction bake-off in Phase 0** — run 2-3 candidates against a small labeled set of your real OMs/rent rolls, score them on the §3a rubric thresholds, and pick on the numbers. That keeps the vendor choice evidence-based and folds into the discovery work already scoped below.

4. Roll-up (de-duped, shared primitives built once)

Scope	Agentic hours	At $100/hr
Discovery / safety plan	24-44	2.4*k*–4.4k
Admin DBMS workstream	190-320	19*k*–32k
Broker segment workstream	95-170	9.5*k*–17k
Listing creation + OM rebuild (first asset class)	260-480	26*k*–48k
Cross-workstream QA / release hardening	40-80	4*k*–8k
Full V1.2 (every listed item, production-grade)	**545-850**	**54.5*k*–85k**
First scoped post-launch tranche	**270-400**	**27*k*–40k**

Adjustments: (1) after reviewing the PDF wizard screenshots the listing-creation workstream rose ~15-20h (explicit Attachments/media step + confirmed asset-class field density); (2) the AI accuracy rubric + evaluation harness (§3a) adds ~14-28h. The rubric harness is front-loaded — it belongs in the first tranche so the AI-accuracy bar is enforced from day one, not retrofitted. The round-number 52k–80k / 26k–38k headline still holds for the bid; these are refinements, not scope expansion.

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5. Recommended phasing

I'd suggest leading with a scoped first tranche rather than a single fixed bid on the whole list. It keeps your spend controlled and lets us validate the new code structure against your live data before you commit the full budget.

Phase 0 — Discovery (1 week) A repo and data-model audit, a production-safety plan, and a tight milestone plan — plus a short document-extraction bake-off (2-3 vendors scored against a labeled sample of your real OMs on the §3a rubric, so the build-vs-buy call in §3c is made on data, not vendor claims). **24-44h → 2.4*k*–4.4k**, credited toward Phase 1 if you proceed.

Phase 1 — First production tranche (260-380h → 26*k*–38k) The highest-value admin improvements, the broker application and approval flow, and one asset class of the listing/OM rebuild — with weekly demos and review points so you can re-prioritize as we go.

Phase 2+ — Remaining V1.2 (balance to 520-800h total) Additional asset classes, full document extraction, OM polish, remaining integrations, production migrations, and hardening — sequenced after Phase 1 proves the structure on your data.

Engagement options — pick whichever fits your budgeting:

- **Hourly at $100/hr with a weekly cap you set** — the lowest-risk option for both of us; you control the burn week to week and can pause anytime.
- **Fixed price per phase** — for example, Phase 1 fixed in the **30*k*–34k** band against a frozen scope, with change-orders for additions. Predictable for budgeting; I carry the velocity risk.

Either way, $100/hr sits at the lower end of the market for senior Rust + Vue contract work, and the AI-native workflow biases delivery toward the low end of each range — so you get senior production quality without paying twice for the migration and QA risk that a live marketplace carries.

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6. Assumptions & open questions

A few things I'd want to confirm with you before locking a fixed price. None of them block starting Phase 0 — they mostly affect where each estimate lands within its range.

1. How many asset classes need to ship in the first release?

2. Is the "separate broker database" a hard **physical** separation, or role/account-segment separation within your existing Postgres system? (A physically separate database adds 16-40h.)
3. Are MailerLite, DocuSign/e-signature, Stripe, BatchData, and your AI provider already integrated in production, with credentials I can access?
4. Would you prefer hard deletes, or soft deletes with audit retention? For transaction records I'd default to soft delete unless you'd rather not.
5. Do you already have approved OM/teaser templates per asset class, or is template/design work in scope?
6. What review do you expect on AI-generated fields before emails/OMs go out to buyers?
7. Which single asset class should ship first? The Listing Creation Steps screens show a hospitality/hotel example — is hotel the pilot, or another class?

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7. Sources

- Feature brief: your V1.2 feature list + Listing Creation Steps PDF
- Platform demo: https://www.youtube.com/watch?v=1Q8RdnQwYLw
- Site: https://reitrades.com/ · Sellers: https://reitrades.com/sellers/ · Buyers: https://reitrades.com/buyers/
- Artur Muller: https://www.linkedin.com/in/mullerartur/
- REItrades company page: https://www.linkedin.com/company/reitradesmarketplace
- REJournals (Artur Muller, 2025-08-20): https://rejournals.com/streamlined-commercial-real-estate-sales-ai-powered-and-automated/